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OneRef: Unified One-tower Expression Grounding and Segmentation with Mask Referring Modeling

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Github: <https://github.com/linhuixiao/OneRef>

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Motivation

Constrained by the separate encoding of vision and language, existing grounding and referring segmentation works heavily rely on bulky Transformer-based fusion en-/decoders and a variety of early-stage interaction technologies. Besides, the current mask visual language modeling (MVLM) fails to capture the nuanced referential relationship between image-text in referring tasks.

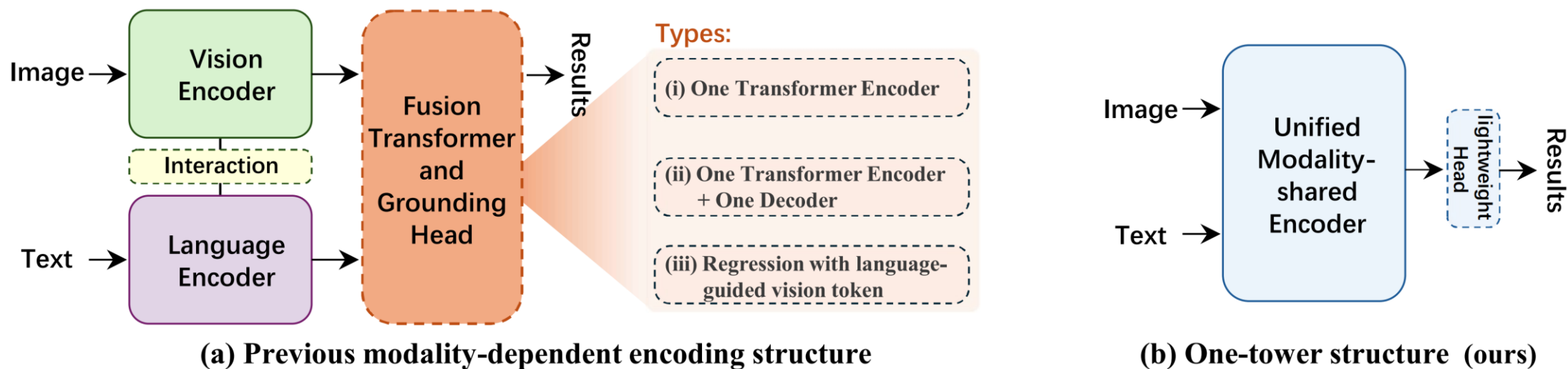


Fig.1 Our OneRef model and previous REC/RES model.

Methodology

We propose *OneRef*, a minimalist referring framework built on the modality-shared one-tower transformer that unifies the visual and linguistic feature spaces. To modeling the referential relationship, we introduce a novel MVLM paradigm called **Mask Referring Modeling (MRefM)**, which encompasses both **referring-aware mask image modeling (Referring MIM)** and **referring-aware mask language modeling (Referring MLM)**.

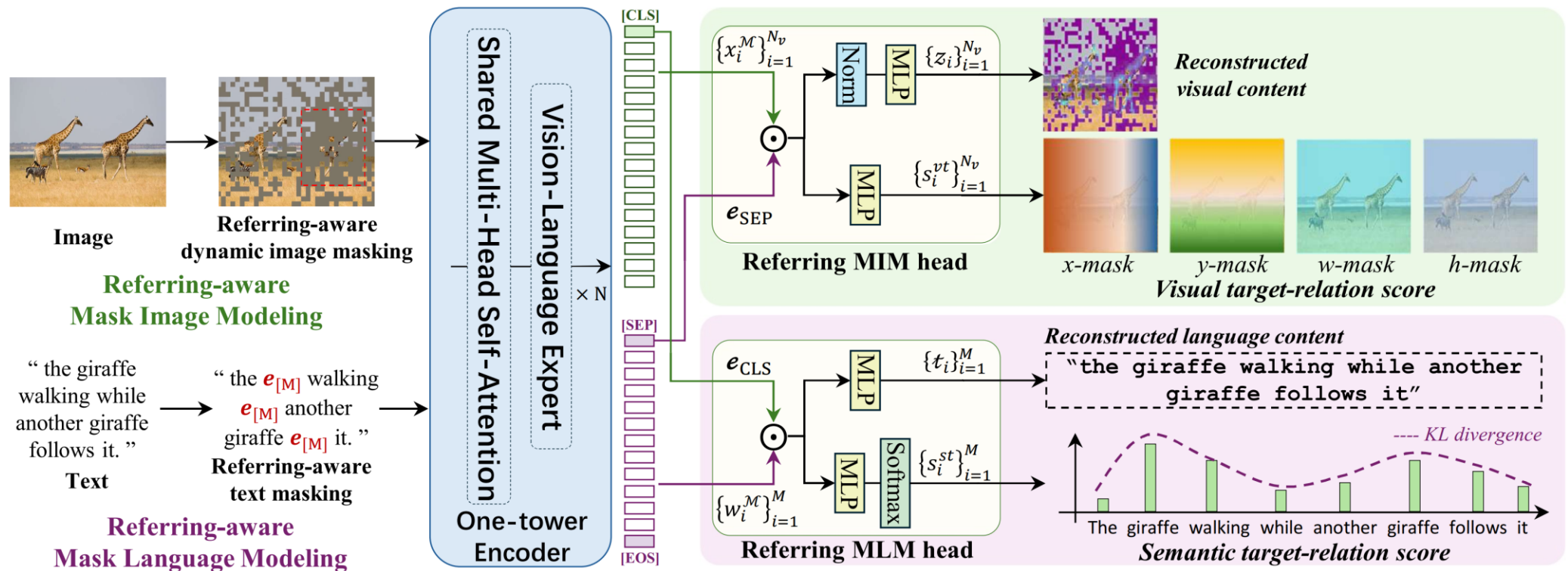


Fig.2 Multimodal Mask Referring Modeling (MRefM) paradigm.

Methodology

Both referring-aware mask image modeling (Referring MIM) and referring-aware mask language modeling (Referring MLM) modules not only reconstruct modality-related content but also cross-modal referring content.

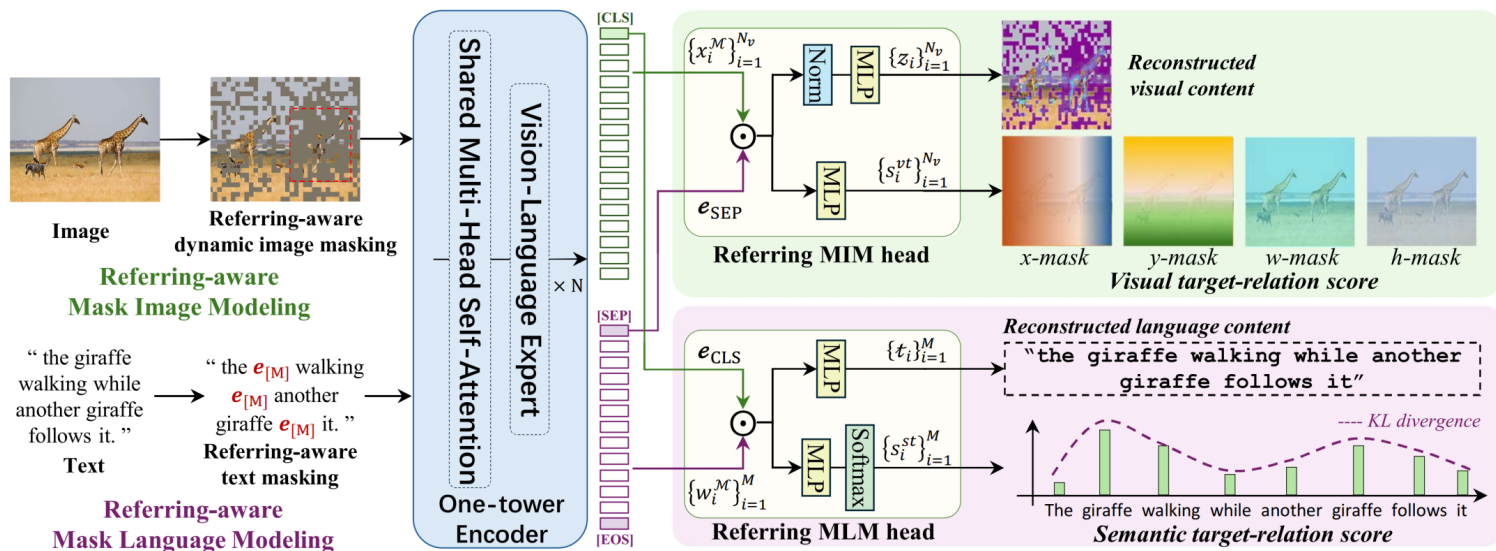


Fig.2 Multimodal Mask Referring Modeling (MRefM) paradigm.

Vanilla MVLM:

$$\mathcal{L}_{\text{MIM}} = - \sum_{\mathbf{x} \in \mathcal{I}} \sum_{i \in \mathcal{M}_v} \log p(\mathbf{z}_i | \mathbf{x}_i^{\mathcal{M}}) \quad \mathcal{L}_{\text{MLM}} = - \sum_{\mathbf{x} \in \mathcal{T}} \sum_{i \in \mathcal{M}_w} \log p(\mathbf{t}_i | \mathbf{w}_i^{\mathcal{M}})$$

Mask Referring Modeling (MRefM):

$$\mathcal{L}_{\text{Referring MIM}} = - \sum_{\mathbf{x} \in \mathcal{I}} \sum_{i \in \mathcal{M}_v} \log p(\mathbf{z}_i | (\mathbf{x}_i^{\mathcal{M}} \odot \mathbf{e}_{\text{SEP}})) - \sum_{\mathbf{x} \in \mathcal{I}} \sum_{i \in [1, N_v]} \log p(\mathbf{s}_i^{vt} | (\mathbf{x}_i^{\mathcal{M}} \odot \mathbf{e}_{\text{SEP}}))$$

$$\mathcal{L}_{\text{Referring MLM}} = - \sum_{\mathbf{w} \in \mathcal{T}} \sum_{i \in \mathcal{M}_w} \log p(\mathbf{t}_i | (\mathbf{w}_i^{\mathcal{M}} \odot \mathbf{e}_{\text{CLS}})) - \sum_{\mathbf{w} \in \mathcal{T}} \sum_{i \in [1, M]} \log p_{kl}(\mathbf{s}_i^{st} | (\mathbf{w}_i^{\mathcal{M}} \odot \mathbf{e}_{\text{CLS}}))$$

Methodology

In **Referring MIM**, we propose referring-aware dynamic image masking strategy with dynamic mask ratio α that is aware of the referred region rather than relying on fixed ratios or generic random masking schemes.

$$\alpha = [\beta \cdot (N_v - N_r) + \gamma \cdot N_r] / N_v.$$

Algorithm 1 Referring-aware Dynamic Masking

Input: N_v image patches, N_r ($h_{rp} \times w_{rp}$) referred patches.

Output: Dynamic masked positions $\mathcal{M}$.

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c  $\leftarrow$  Rand Select  $\beta \cdot N_v$  numbers in  $[1, N_v]$ 
New  $\mathcal{M} \in \mathbb{R}^{1 \times N_v}$ ,  $\{\{\mathcal{M}_i\}_i^{N_v} \mid \mathcal{M}_i = 1 \text{ if } i \in \mathbf{c}, \text{ else } 0\}$ 
 $\mathcal{M} \leftarrow \mathcal{M}$  reshape as  $\mathcal{M} \in \mathbb{R}^{h \times w}$   $\triangleright$  In-context masking
New  $\mathcal{M}_r \in \mathbb{R}^{h_{rp} \times w_{rp}}$  with all as 0  $\triangleright$  Referred masking
while  $|\mathcal{M}_r| \leq \gamma \cdot N_r$  do
     $s \leftarrow \text{Rand}(1, \gamma \cdot N_r - |\mathcal{M}_r|)$   $\triangleright$  Block size
     $r \leftarrow \text{Rand}(a, \frac{1}{a})$   $\triangleright$  Aspect ratio of block
     $w_b \leftarrow \sqrt{s/r}; h_b \leftarrow \sqrt{s \cdot r}$   $\triangleright$  Width, height of block
     $l \leftarrow \text{Rand}(0, w_{rp} - w_b); t \leftarrow \text{Rand}(0, h_{rp} - h_b)$ 
     $\{\mathcal{M}_r(i, j) = 1 \mid i \in [l, l + w_b), j \in [t, t + h_b)\}$ 
end
 $\mathcal{M}(x_{sp} : x_{sp} + w_{rp}, y_{sp} : y_{sp} + h_{rp}) = \mathcal{M}_r$ 
return  $\mathcal{M}$ .
  
```

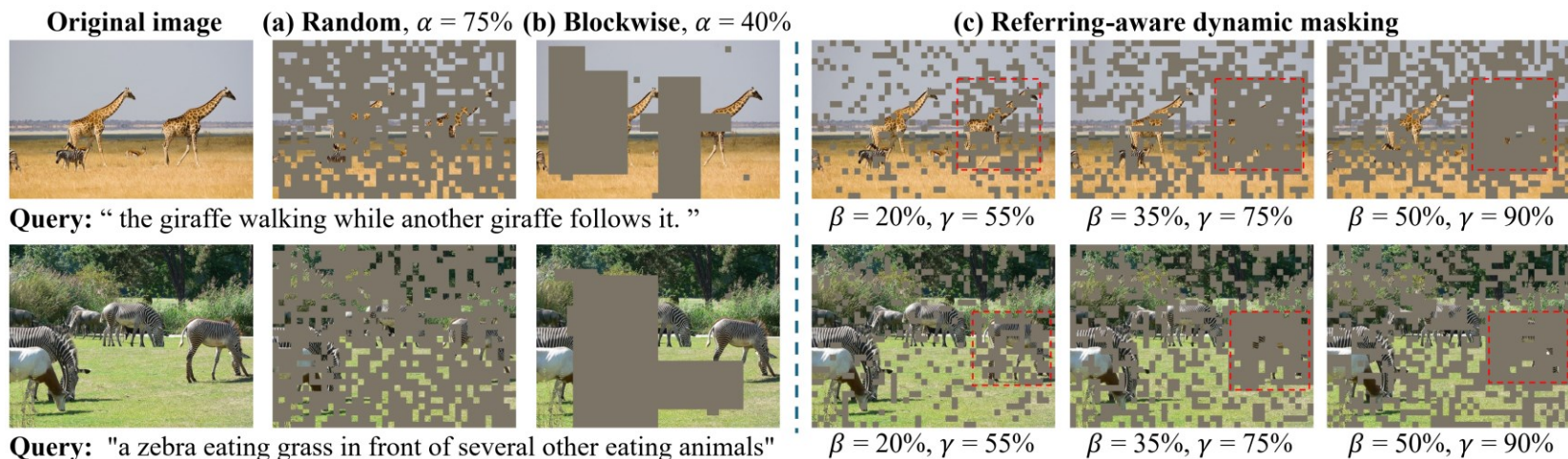


Fig.3 Our proposed referring-aware dynamic masking.

In **Referring MIM**, the similarity between vocabularies and referred region are obtained as:

$$\mathbf{s}^{st} = \lambda_{reg} \cdot \sigma(< \mathbf{e}_{CLS}^{reg \top}, \{\mathbf{w}_i\}_{i=1}^M >) + \lambda_{img} \cdot \sigma(< \mathbf{e}_{CLS}^{img \top}, \{\mathbf{w}_i\}_{i=1}^M >)$$

Methodology

By leveraging the unified visual language feature space and incorporating MRefM's ability to model the referential relations, our approach enables direct regression of the referring results without resorting to various complex techniques.

$$\hat{\mathcal{B}} = \text{MLP}\left(\sum_{i \in [1, N_v]} (\text{Repeat}(\sigma(\langle \mathbf{e}_{\text{ESP}}^\top, \{\mathbf{x}_i\}_{i=1}^{N_v} \rangle)) \odot \text{MLP}(\{\mathbf{x}_i\}_{i=1}^{N_v}))\right).$$

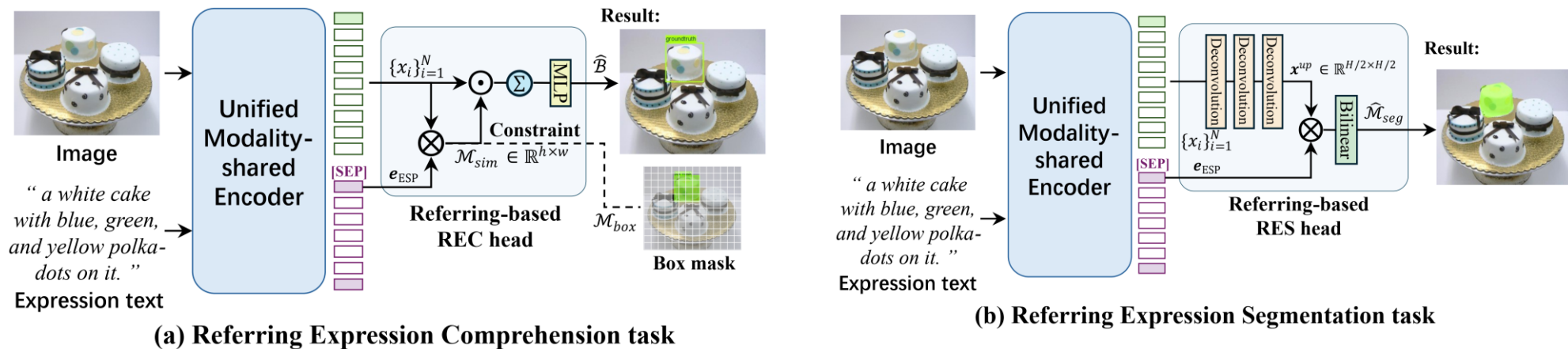


Fig.4 Referring-based grounding and segmentation transfer.

Training Objectives for REC and RES transfer:

REC: $\mathcal{L}_{REC} = \lambda_{L_1} \mathcal{L}_{L_1}(\hat{\mathcal{B}}, \mathcal{B}) + \lambda_{giou} \mathcal{L}_{giou}(\hat{\mathcal{B}}, \mathcal{B}) + \mathcal{L}_{box_mask_constraints}$

$$\mathcal{L}_{box_mask_constraints} = \lambda_{f_box} \mathcal{L}_{focal}(\mathcal{M}_{sim}, \mathcal{M}_{box}) + \lambda_{d_box} \mathcal{L}_{dice}(\mathcal{M}_{sim}, \mathcal{M}_{box}).$$

RES: $\mathcal{L}_{RES} = \lambda_{f_seg} \mathcal{L}_{focal}(\hat{\mathcal{M}}_{seg}, \mathcal{M}_{seg}) + \lambda_{d_seg} \mathcal{L}_{dice}(\hat{\mathcal{M}}_{seg}, \mathcal{M}_{seg})$

Results

Our method consistently surpasses existing approaches and achieves SoTA performance on both grounding and segmentation tasks (Tab. 1, 2, 3), providing valuable insights for future research.

Tab.1 Comparison with SoTA methods on REC task (Acc@0.5 metric).

Methods	Venue	Visual Backbone	Language Backbone	RefCOCO			RefCOCO+			RefCOCOg		ReferIt test	Flickr test
				val	testA	testB	val	testA	testB	val	test		
Single-dataset fine-tuning setting w. uni-modal pre-trained close-set detector and language model: (traditional setting)													
TransVG [14]	ICCV'21	RN101+DETR	BERT-B	81.02	82.72	78.35	64.82	70.70	56.94	68.67	67.73	70.73	79.10
Word2Pix [103]	TNNLS'22	RN101+DETR	BERT-B	81.20	84.39	78.12	69.74	76.11	61.24	70.81	71.34	–	–
QRNet [98]	CVPR'22	Swin-S [60]	BERT-B	84.01	85.85	82.34	72.94	76.17	63.81	71.89	73.03	74.61	81.95
VG-LAW [79]	CVPR'23	ViT-Det [46]	BERT-B	86.06	88.56	82.87	75.74	80.32	66.69	75.31	75.95	76.60	–
TransVG++[15]	TPAMI'23	ViT-Det [46]	BERT-B	86.28	88.37	80.97	75.39	80.45	66.28	76.18	76.30	74.70	81.49
Single-dataset fine-tuning setting w. vision-language self-supervised pre-trained model:													
CLIP-VG [91]	TMM'23	CLIP-B	CLIP-B	84.29	87.76	78.43	69.55	77.33	57.62	73.18	72.54	70.89	81.99
JMRI [108]	TIM'23	CLIP-B	CLIP-B	82.97	87.30	74.62	71.17	79.82	57.01	71.96	72.04	68.23	79.90
Dynamic-MDETR	TPAMI'23	CLIP-B	CLIP-B	85.97	88.82	80.12	74.83	81.70	63.44	74.14	74.49	70.37	81.89
HiVG-B [92]	ACMMM'24	CLIP-B	CLIP-B	87.32	89.86	83.27	78.06	83.81	68.11	78.29	78.79	75.22	82.11
HiVG-L [92]	ACMMM'24	CLIP-L	CLIP-L	88.14	91.09	83.71	80.10	86.77	70.53	80.78	80.25	76.23	82.16
OneRef-B (ours)	NeurIPS'24	BEiT3-B	BEiT3-B	88.75	90.95	85.34	80.43	86.46	74.26	83.68	83.52	77.17	83.61
OneRef-L (ours)	NeurIPS'24	BEiT3-L	BEiT3-L	92.87	94.01	90.19	87.98	91.57	83.73	88.11	89.29	81.11	84.75

Results

Our method consistently surpasses existing approaches and achieves SoTA performance on both grounding and segmentation tasks (Tab. 1, 2, 3), providing valuable insights for future research.

Tab.2 Comparison with SoTA methods on REC task (Acc@0.5 metric).

Methods	Venue	Visual/Language Backbone	Intermediate pretrain data	Data size	RefCOCO			RefCOCO+			RefCOCOg	
					val	testA	testB	val	testA	testB	val	test
Dataset-mixed intermediate pre-training setting (w. box-level dataset-mixed open-set detection pre-trained model)												
MDETR [†] [33]	ICCV'21	RN101/RoBERT-B	GoldG,RefC	6.5M	86.75	89.58	81.41	79.52	84.09	70.62	81.64	80.89
YORO [†] [29]	ECCV'22	ViLT [37] / BERT-B	GoldG,RefC	6.5M	82.90	85.60	77.40	73.50	78.60	64.90	73.40	74.30
DQ-DETR [†] [54]	AAAI'23	RN101 / BERT-B	GoldG,RefC	6.5M	88.63	91.04	83.51	81.66	86.15	73.21	82.76	83.44
Grounding-DINO-B [†]	arXiv'23	Swin-T / BERT-B	O365,GoldG,RefC	7.2M	89.19	91.86	85.99	81.09	87.40	74.71	84.15	84.94
Grounding-DINO-L [†]	arXiv'23	Swin-L / BERT-B	G-DINO-L*	21.4M	90.56	93.19	88.24	82.75	88.95	75.92	86.13	87.02
CyCo [84]	AAAI'24	ViT[19]/ BERT-B	VG,SBU,CC3M, <i>etc.</i>	>120M	89.47	91.87	85.33	80.40	87.07	69.87	81.31	81.04
HiVG-B [†] [92]	ACMMM'24	CLIP-B / CLIP-B	RefC,ReferIt,Flickr	0.8M	90.56	92.55	87.23	83.08	87.83	76.68	84.71	84.69
HiVG-L [†] [92]	ACMMM'24	CLIP-L / CLIP-L	RefC,ReferIt,Flickr	0.8M	91.37	93.64	88.03	83.63	88.16	77.37	86.73	86.86
Fine-tuning setting w. dataset-mixed multi-task mix-supervised pre-trained model:												
UniTAB [†] [97]	ECCV'22	RN101/RoBERT-B	VG,COCO, <i>etc.</i>	>20M	88.59	91.06	83.75	80.97	85.36	71.55	84.58	84.70
OFA-B [†] [85]	ICML'22	OFA-B / OFA-B	–	–	88.48	90.67	83.30	81.39	87.15	74.29	82.29	82.31
OFA-L [†] [85]	ICML'22	OFA-L / OFA-L	–	–	90.05	92.93	85.26	85.80	89.87	79.22	85.89	86.55
Fine-tuning setting w. grounding multimodal large language model (GMLLM):												
Shikra-7B [†] [10]	arXiv'23	CLIP-L / Vicuna-7B[12]	RefC,VG	0.5M	87.01	90.61	80.24	81.60	87.36	72.12	82.27	82.19
Ferret-7B [†] [100]	ICLR'24	CLIP-L / Vicuna-7B[12]	GRIT [100]	>8M	87.49	91.35	82.45	80.78	87.38	73.14	83.93	84.76
LION-4B [†] [9]	CVPR'24	EVA-G[21]/FlanT5-3B	VG,COCO, <i>etc.</i>	3.6M	89.73	92.29	84.82	83.60	88.72	77.34	85.69	85.63
LION-12B [†] [9]	CVPR'24	EVA-G[21]/FlanT5-11B	VG,COCO, <i>etc.</i>	3.6M	89.80	93.02	85.57	83.95	89.22	78.06	85.52	85.74
OneRef-B [†] (unsupervised)		BEiT3-B / BEiT3-B	RefC,ReferIt	0.5M	89.16	92.03	87.26	83.18	88.56	77.66	84.72	85.17
OneRef-B [†] (0.2B)	NeurIPS'24	BEiT3-B / BEiT3-B	RefC,ReferIt	0.5M	91.89	94.31	88.58	86.38	90.38	79.47	86.82	87.32
OneRef-L [†] (0.6B)	NeurIPS'24	BEiT3-L / BEiT3-L	RefC,ReferIt	0.5M	93.21	95.43	90.11	88.35	92.11	82.70	87.81	88.83

Results

Our method consistently surpasses existing approaches and achieves SoTA performance on both grounding and segmentation tasks (Tab. 1, 2, 3), providing valuable insights for future research.

Tab.3 Comparison with SoTA methods on RES task (mIOU metric).

Methods	Venue	Visual/Language Backbone	Intermediate pretrain data	RefCOCO			RefCOCO+			RefCOCOg	
				val	testA	testB	val	testA	testB	val	test
Single-dataset fine-tuning setting w. uni-modal pre-trained close-set segmentation model: (traditional setting)											
RefTR [45]	NIPS'21	RN101 / BERT-B	–	74.34	76.77	70.87	66.75	70.58	59.40	66.63	67.39
SeqTR [107]	ECCV'22	DN53[72]/Bi-GRU	–	71.70	73.31	69.82	63.04	66.73	58.97	64.69	65.74
LAVT [94]	CVPR'22	Swin-B / BERT-B	–	74.46	76.89	70.94	65.81	70.97	59.23	63.34	63.62
VG-LAW [79]	CVPR'23	ViT-Det / BERT-B	–	75.05	77.36	71.69	66.61	70.30	58.14	65.36	65.13
Single-dataset fine-tuning setting w. vision-language self-supervised pre-trained model:											
CRIS [89]	CVPR'22	CLIP-L / CLIP-L	–	70.47	73.18	66.10	62.27	68.06	53.68	59.87	60.36
JMCELN[32]	EMNLP'23	CLIP-B / CLIP-B	–	74.40	77.69	70.43	66.99	72.69	57.34	64.08	64.99
RISCLIP-B [35]	NAACL'24	CLIP-B / CLIP-B	–	75.68	78.01	72.46	69.16	73.53	60.68	67.62	67.97
RISCLIP-L [35]	NAACL'24	CLIP-L / CLIP-L	–	78.87	81.46	75.41	74.38	78.77	66.84	71.82	71.65
OneRef-B (ours)	NeurIPS'24	BEiT3-B / BEiT3-B	–	77.57	79.05	75.11	71.25	75.41	65.45	69.37	69.70
OneRef-L (ours)	NeurIPS'24	BEiT3-L / BEiT3-L	–	80.09	82.19	77.51	75.17	79.38	70.17	73.18	72.76
Dataset-mixed intermediate pre-training setting:											
PolyFormer-B [†] [53]	CVPR'23	Swin-B / BERT-B	RefC	75.96	77.09	73.22	70.65	74.51	64.64	69.36	69.88
RISCLIP-B [†] [35]	NAACL'24	CLIP-B / CLIP-B	RefC	75.68	78.01	72.46	72.46	74.30	61.37	69.49	69.53
RISCLIP-L [†] [35]	NAACL'24	CLIP-L / CLIP-L	RefC	79.53	81.78	75.78	74.88	78.88	68.09	73.45	74.52
OneRef-B [†] (unsupervised)		BEiT3-B / BEiT3-B	RefC	78.20	79.26	75.92	72.54	75.54	67.39	71.28	71.13
OneRef-B [†] (ours)	NeurIPS'24	BEiT3-B / BEiT3-B	RefC	79.83	81.86	76.99	74.68	77.90	69.58	74.06	74.92
OneRef-L [†] (ours)	NeurIPS'24	BEiT3-L / BEiT3-L	RefC	81.26	83.06	79.45	76.60	80.16	72.95	75.68	76.82

Results

Our method consistently surpasses existing approaches and achieves SoTA performance on both grounding and segmentation tasks (Tab. 1, 2, 3), providing valuable insights for future research.

Query text : “white tourist ship closest viewer. ”



Query text: “a decorated passenger ship is waiting for departure in a port. ”



Query text: “woman with blond hair. ”



Query text: “a blue shirt person giving presentation. ”



Original image

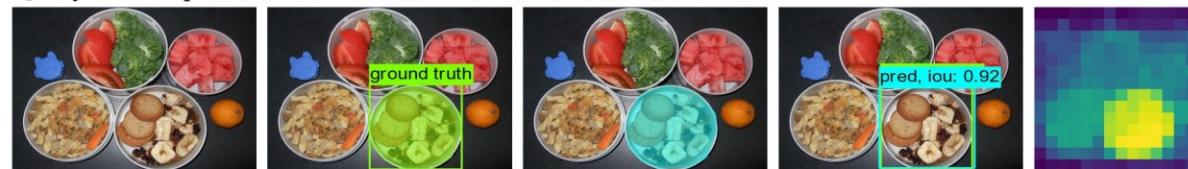
Ground truth

RES Prediction

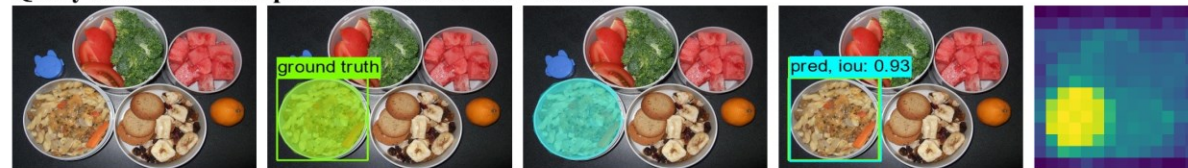
REC Prediction

Cross-modal feature

Query text: “a plate of cookies and banana slice , raisins , and craisins. ”



Query text: “a bowl of pasta next to a bowl of salad. ”



Query text: “a group of six oranges on a wooden table surrounded by several other fruits and vegetables. ”



Query text: “the bag of vegetables behind all of the other fruits and vegetables. ”



Original image

Ground truth

RES Prediction

REC Prediction

Cross-modal feature

Fig.5 Part of the qualitative results in REC and RES tasks.

Conclusion

We propose a novel, highly concise, and feature space unified one-tower referring framework. Additionally, we pioneer the exploration of mask modeling in referring tasks by introducing MRefM paradigm to capture the referential relationships between vision and text. MRefM enables potential large-scale pre-training of grounding in future, presenting a new direction for referring tasks.



Paper



Code



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